Assets, Rents, and the Socialized Buildout

An accounting of the AI capital cycle: market values, capital stocks, revenue requirements, the buildout precedents, and the July 2026 rotation

Prepared by Claude (Anthropic) from a dialogue with Matthew Tomei · Second edition, August 4, 2026

Abstract. Roughly $27 trillion of equity value has attached to AI-related companies since November 2022, against which the physically deployed, depreciation-adjusted AI capital stock is on the order of $1.2 trillion. This report separates the two quantities and sizes what must fill the gap. It builds a vintage model of the AI capital stock, reprices it at competitive supply-chain margins, backs out the economic-rent stream the market has capitalized, and inverts that requirement into the share of world GDP that must flow to AI at plausible margins. It calibrates the requirement against forty years of IT-spending history; develops three completed buildouts - Britain's Railway Mania, the American railroads, and telecom 1996-2002 - as structural precedents, covertly socialized infrastructure programs whose losses funded a commons; introduces the toll-booth layer that historically kept the profits the network capital never earned; maps the related-party pathologies of 1873 and 1996-2002 (Credit Mobilier; Lucent, Nortel, Winstar) onto today's circular deal graph; and reads the July 2026 rotation - the forced unwind of Situational Awareness LP and Microsoft's $480 billion single-session repricing - as the first live experiment on where AI's rents will land. Reported figures and the authors' estimates are distinguished throughout; sources and model assumptions appear at the end.

1. The Question and a Correction

The prompt for this analysis was a Goldman Sachs Insights piece (July 2026) asking whether US equity valuations have outrun fundamentals [1]. Its load-bearing figures: AI-related companies have added roughly $27 trillion of market value since November 2022 (up from about $19 trillion seven months earlier), while the bank's baseline estimate of the present discounted value of AI-related capital revenues to US companies is roughly $9 trillion. US technology investment as a share of GDP has surpassed its late-1990s peak, and the 2026 spending plans of the largest cloud and computing companies came in about fifty percent higher than estimates made only six months prior. Goldman's stated risk is that the market overestimates the persistence of the earnings streams - particularly for the companies supplying the capex boom - beyond a two-to-three-year horizon, and it concedes the reflexive point that the investment boom itself currently generates a substantial portion of the profits justifying the prices.

One correction frames everything downstream: the $27 trillion is not investment. It is market-capitalization appreciation - capitalized expectation. Actual capital expenditure is an order of magnitude smaller. The four largest hyperscalers spent about $226 billion in 2024 and about $410 billion in 2025, and after Q1 2026 earnings guided to roughly $725 billion combined for 2026, with Microsoft alone at $190 billion [13, 15]. Goldman projects $5.3 trillion of big-four capex over 2025-2030 [1]. Cumulative AI-attributable investment since ChatGPT, built up carefully in Section 3, is roughly $1.0-1.2 trillion gross. Realized investment is therefore about six percent of the market-value move; the other ninety-four percent is expectation. That ratio is the article's tension stated as arithmetic, and the remainder of this report is an attempt to give each side of the ratio its correct accounting.

2. The Capex Boom and Its Persistence

The mechanical core of the persistence concern is that supplier revenue is a derivative of the capex flow, not of the installed stock. If spending merely plateaus, supplier growth goes to zero; if it reverts toward replacement level, supplier revenue contracts outright. The historical base rate is unkind - US telecom capex roughly halved between 2000 and 2003 - but the depreciation structure here genuinely differs from fiber. Dark fiber carried near-zero replacement demand; accelerators on four-to-six-year lives at a ~$500 billion annual run rate imply a replacement floor above $100 billion per year and rising with the installed base. Short asset lives are bearish for the buyers' balance sheets and bullish for recurring supplier revenue: the same fact, booked on opposite sides.

The stronger evidence for the limited-time view is the financing signature. Capital intensity now runs 45-57 percent of revenue for the big four and 86 percent for Oracle; Amazon's free cash flow is projected to turn negative; roughly $108 billion of debt was raised by hyperscalers in 2025 with on the order of $1.5 trillion projected over the coming years [15]. Moody's now projects $785 billion of capital expenditure by the six largest US hyperscalers in 2026, approaching $1 trillion in 2027, part-funded by roughly $175 billion of debt issuance this year [28, 35]. Migration from cash-funded to debt-and-lease-funded spending is the canonical late-phase marker: it extends the boom past internal cash generation and makes the flow fragile to sentiment.

2.1 Production capacity: reallocation, not new build

A second-order question with first-order consequences: how much of the apparent explosion in compute production is new physical capacity versus existing capacity redirected and repriced? So far, mostly the latter. Leading-edge logic barely grew in aggregate: high-performance computing reached 61 percent of TSMC revenue by Q1 2026, up from roughly a third before the boom - the same fabs, shifted from phones to accelerators as mobile demand softened [14]. Memory is the purest relabeling: HBM consumed 23 percent of global DRAM wafer starts in 2026 versus 8 percent in 2024, at roughly a three-to-one wafer-intensity penalty against DDR5; Micron exited consumer memory entirely, and DDR5 contract pricing more than doubled [16]. The one place physical capacity genuinely multiplied is advanced packaging, because it was the binding constraint and is cheap relative to fabs: TSMC CoWoS went from about 35,000 wafers per month at end-2024 to about 75,000 at end-2025, targeting 125,000-130,000 by end-2026 - a near-4x expansion that still trails demand by an estimated 10-20 percent [16]. Meanwhile TSMC's entire 2026 capital budget is $52-56 billion and total semiconductor-industry capex runs about $200 billion per year - against more than $725 billion of downstream datacenter spending. The production base for compute is expanding far more slowly than spending on its output.

That asymmetry concentrates pricing power upstream - TSMC at a 66.2 percent gross margin, memory prices doubling - and Microsoft attributed $25 billion of its 2026 capex increase purely to rising memory and component costs [13, 14]. A nontrivial slice of measured capex growth is therefore intra-supply-chain price inflation: scarcity rents, which are definitionally transient because they persist only while supply lags. The market capitalizing scarcity rents as durable franchise earnings is the mechanically precise form of Goldman's earnings-bubble worry. The watch variables are the CoWoS supply gap narrowing toward ~10 percent through 2026 and the new memory fabs landing in 2027-28 - fresh supply arriving just as capex growth must decelerate is the classic semiconductor bullwhip setup. The falsifier runs through end demand: cloud AI revenue compounding at rates like Google Cloud's reported 63 percent year-over-year is the kind of number that, sustained, absorbs the supply as it lands [13].

3. An Asset-Side Estimate: The AI Capital Stock

The direct route around speculative pricing is to build the asset side: a replacement-cost capital stock, compared against market value - Tobin's Q at sector level. Q above one is not per se irrational, since market value legitimately capitalizes rents on future investment as well as installed assets; the exercise does not eliminate the speculative component but quarantines it into a single residual that can be sized and argued about. Aggregate versions exist - Econbrowser ran whole-market q against the intellectual-property capital stock in December 2025 [17] - and Goldman's $9 trillion is the profit-side dual, but an AI-sector decomposition does not appear to have been published. We construct it here.

3.1 Vintage build-up

Global AI-attributable gross capex by vintage, estimated from hyperscaler disclosures (applying an AI share rising from roughly 30 percent in 2023 to 75 percent in 2026, per CreditSights [15]) plus Oracle, the neoclouds, xAI and OpenAI's direct spending, Chinese platforms, and sovereign programs:

Vintage	Low ($B)	Central ($B)	High ($B)	Age at mid-2026 (yr)
2023	45	65	85	3.0
2024	150	185	215	2.0
2025	400	460	500	1.0
H1 2026	310	370	420	0.25
Cumulative gross	905	1,080	1,220	-

Table 1. Vintage gross AI-attributable capex (authors' estimates from reported figures).

A supply-side cross-check anchors the range. Nvidia's datacenter revenue was $115.2 billion in FY2025, $193.7 billion in FY2026, and $75.2 billion in Q1 FY2027 alone - cumulative roughly $430 billion from early 2023 through April 2026, approaching $500 billion through July [12]. At Nvidia around 60 percent of AI IT-equipment value (custom accelerators such as TPU and Trainium at cost, AMD, non-Nvidia networking making up the rest), implied IT spending of $750-830 billion pushes the true gross figure toward the upper half of the modeled range.

3.2 Depreciation, add-ons, and the competitive-cost adjustment

Depreciating by asset class - 62 percent IT equipment on five-year straight line, 23 percent power and cooling infrastructure on fifteen, 15 percent shell and land on thirty, mid-year convention - removes roughly $160 billion. To the net datacenter stock we add the AI-allocated share of upstream fab plant (TSMC and the memory makers, ~$125B), AI-attributable grid and power investment (~$65B), and model intellectual property at cost, excluding compute already counted (~$55B). One further adjustment matters: book value embeds the scarcity rents, because the accelerator stock was purchased at ~75 percent gross margins and memory at doubled prices. Repricing the IT stock at competitive supply-chain margins cuts it roughly in half, giving the long-run equilibrium anchor that entry drives toward - and implying the asset base itself deflates by roughly $250 billion if supplier pricing normalizes.

The five-year IT life deserves defense, because the industry's own schedules mostly say six. Between fiscal 2021 and 2025 the major operators repeatedly extended server useful lives, adding non-cash operating income of roughly $3.7 billion (Microsoft, FY2023, to six years), $3.9 billion (Alphabet, 2023), $3.2 billion (Amazon, 2024), $2.9 billion (Meta, 2025), and $0.7 billion (Oracle, FY2025) - more than $14 billion per year of reported operating income, roughly $11.7 billion after tax, manufactured across the complex by an assumption about how long a chip stays useful, per the companies' own change-in-estimate disclosures [33]. The reversal is the honest signal: effective January 2025, Amazon shortened a subset of server and networking lives from six years back to five, citing in its 10-K the increased pace of AI and machine-learning hardware development, at a cost of $1.4 billion of additional depreciation in 2025 [26, 33]. The operator with the longest experience running datacenters at scale agrees with this model's assumption against its peers' schedules.

Case	Gross capex	Net DC stock	Total net stock*	Competitive-cost stock
Low	905	773	958	746
Central	1,080	919	1,164	911
High	1,220	1,034	1,339	1,056

*Table 2. AI capital stock at mid-2026, $B. *Includes upstream fabs, power/grid, and model IP at cost. Authors' model; assumptions in Appendix A.*

3.3 The rent decomposition

Against the $27 trillion market-value gain - or roughly $18 trillion taking Goldman's own caveat that perhaps a third reflects non-AI businesses - sector Q runs 15-23x. The rent capitalization (market value minus assets) is about $17 trillion on the conservative attribution. Backed out as a required flow at an 8 percent discount rate:

AI-attributable MV	Rent capitalization	Growing 3%/yr	Flat perpetuity	Decaying 10%/yr
$14T	$12.8T	$0.64T/yr	$1.03T/yr	$2.31T/yr
$18T	$16.8T	$0.84T/yr	$1.35T/yr	$3.03T/yr
$27T	$25.8T	$1.29T/yr	$2.07T/yr	$4.65T/yr

Table 3. Required economic-rent flow implied by rent capitalization at r = 8%, central asset stock of $1.18T. Authors' model.

Current realized AI economic rents run roughly $330-350 billion per year: Nvidia annualizing about $200 billion of operating income, TSMC's AI-attributable share around $40 billion, memory scarcity profits around $60 billion, thin hyperscaler direct-AI margin, and an application layer that does not earn what its software analogues did [12, 14, 37]. The market therefore requires the rent stream to grow roughly 2.5-4x and persist indefinitely - while 85-90 percent of today's rents sit in the supply chain, in exactly the scarcity-rent form that mean-reverts. The reconciliation with Goldman is clean: $1.2 trillion of assets plus their $9 trillion rent PDV accounts for about $10 trillion against $18 trillion of AI-attributable market gain, leaving roughly $8 trillion as the speculative residual under their own baseline. A fully auditable variant exists for anyone who wants it: summing net property, plant and equipment plus finance-lease assets from the 10-Ks of the AI complex and taking the delta since Q4 2022 yields roughly $450-500 billion for the big four through 2025, corroborating the vintage build once non-AI growth is stripped and the rest of the ecosystem added.

The model layer needs a note, because the margin figures that circulate for it are not comparable to each other. DeepSeek's self-published 545 percent and the roughly 70 percent “compute margin” reported for OpenAI are ratios of price to marginal serving cost, and on that measure serving tokens is genuinely cheap: DeepSeek's own disclosure put a day of V3/R1 inference at $87,072 of GPU rental against $562,027 of theoretically billable output, while noting that its web and app tiers are free, that off-peak pricing is discounted, and that realized revenue was far lower [37]. Blended gross margin is a different quantity and a much smaller one. OpenAI's adjusted gross margin ran about 33 percent in 2025, down from 40 percent and below its own 46 percent plan; Anthropic's 2025 guidance was cut by roughly ten points to about 40 percent when inference rented from Google and Amazon overran plan by about a quarter [37]. Free tiers, flat-rate pricing over metered compute, training amortization, and rented capacity all sit between the marginal cost and the income statement - and flat rates invert the software margin logic, because the heaviest users are the least profitable, which is why the $200-per-month tier lost money by its own seller's account. At the application layer the same wedge shows up as a gross-margin gap of twenty-five to thirty-five points against traditional software - 33 percent in 2024 rising to a projected 45 percent in 2026, against 75-85 percent for SaaS - with flagship AI-native firms running negative gross margin at billion-dollar revenue scale, and turning positive by integrating backward into their own models [37]. That last move is the tell: the margin is recoverable by whoever owns the inference, which is the rent-location question of Section 4.1 posed one layer down.

3.4 Water: a nineteenth-century precedent for the method

The method has a precedent, run by contemporaries on the railroads. Total railroad capitalization grew from $4.6 billion in 1876 to $10.6 billion by 1890, and one railroad historian estimated that forty percent of the latter figure was “water” - securities issued in excess of any investment in roadbed, rails, or rolling stock, floated against branch lines not yet built and towns that existed only on promotional maps [29]. Water is precisely a replacement-cost gap: market value minus tangible assets, named and measured a century before Tobin. The railroad sector's implied Q at peak watering was roughly 1.7x. The AI complex's is 15-23x. The honest reading of that spread is a fork: either genuine intangibles - the models, the know-how, the network positions - are worth an order of magnitude more relative to the physical plant than anything in 1890, or the watering is proportionally larger. Naming which of those one believes is the entire argument.

4. The Revenue Requirement: What Share of GDP Must Flow to AI

Inverting the rent requirement into a revenue requirement demands one construction choice: AI revenue must be measured as external revenue - sales from the consolidated AI stack to the rest of the economy. Summing layer revenues double-counts (Nvidia's revenue is Microsoft's cost is OpenAI's cost), and today most of the stack's profit is funded by intra-stack capex - by capital markets rather than final demand. The transition being priced is precisely that external revenue replaces the capex flow as the funding source of the rents.

Required after-tax profit by roughly 2030 is the rent stream plus normal returns on what will by then be a ~$5 trillion capital stock: about $1.25 trillion per year in the growing-rents case, $1.75 trillion flat, $3.4 trillion if rents decay. Dividing by consolidated net margin on external revenue:

Case (required profit)	30% margin	20% margin	12% margin
Growing rents ($1.25T/yr)	$4.1T = 3.1% W / 12% US	$6.2T = 4.6% W / 18% US	$10.4T = 7.7% W / 30% US
Flat perpetuity ($1.75T/yr)	$5.8T = 4.3% W / 17% US	$8.8T = 6.5% W / 25% US	$14.6T = 10.8% W / 42% US
Decaying rents ($3.4T/yr)	$11.4T = 8.5% W	$17.2T = 12.7% W	$28.6T = 21.2% W

Table 4. Required external AI revenue and share of 2030 world (~$135T) and US (~$35T) GDP. Authors' model; Appendix B.

Three readings. First, the decay row is the valuation restated as a theorem: the pricing requires durable moats, because without them no plausible GDP share closes the gap. Second, the US-only column is arithmetic nonsense at 12-42 percent of GDP, so the valuations mathematically presuppose global revenue capture - which sits awkwardly against export controls and Chinese self-supply. Third, the bracket: the floor - merely making the buildout NPV-neutral, covering depreciation (~$940B/yr on the 2030 stock, dominated by five-year IT lives), operations, and cost of capital with zero rents - is about $1.7 trillion per year, 1.3 percent of world GDP. Current external AI revenue (cloud AI services, the application layer, enterprise on-premise; authors' estimate) runs perhaps $300-400 billion, about 0.3 percent of world GDP. The sector needs roughly 4-5x revenue growth to stop destroying value and 12-18x to justify the market pricing. External triangulations land on the same gap: Bain estimates roughly $2 trillion of annual AI revenue required by 2030 and finds the world about $800 billion short even if every on-premise IT dollar migrates to cloud and every productivity saving is reinvested [36], while Sequoia's David Cahn has walked his running estimate of the revenue question from $200 billion to $600 billion to roughly $840 billion [28].

For scale: 3-6.5 percent of world GDP means external AI spending matching or exceeding all current global IT spending and two to three times all software plus cloud combined, and $1.25-1.75 trillion of after-tax profit stands against a total US corporate-profit pool of roughly $3.9 trillion. Which raises the question the GDP framing usually obscures: which spending pool funds this? IT budgets cannot - they total about five percent of world GDP and AI will not take all of them. Consumer discretionary does not produce trillions at software margins. The only pool of sufficient size is labor compensation, roughly 53 percent of GDP, about $70 trillion by 2030. If AI vendors capture around 30 percent of the labor-cost savings they create, then $4-7 trillion of revenue implies AI substantively addressing 19-33 percent of global labor income. The percentage-of-GDP assumption thus decomposes into four separately contestable conjuncts: partial automation of a fifth to a third of world labor; the AI complex capturing about 30 percent of that surplus rather than competing it away to users; doing so at 25-30 percent consolidated margins; and moats durable enough that the rents do not decay. The margin assumption and the moat assumption are the same assumption in different clothes. Telecom built the internet at ten percent margins and never earned its cost of capital; software is the other precedent. The market has priced the software outcome.

4.1 The rent-location corollary

Everything above concerns the level of the required rent stream and says nothing about its distribution - and history's answer to the distribution question inverts the market's. In each precedent of Section 6, the capital that built the network earned poor returns while asset-light businesses riding the network - owning the customer relationship, the standard, or the validated record, at near-zero incremental capital cost - kept the profit: the express companies and Pullman on the railroads; Google, Amazon, and Netflix on fiber that bankruptcy had made nearly free. The market's current allocation is the opposite: the rent capitalization sits overwhelmingly in the builders and suppliers, while incumbent application software entered mid-2026 trading at 11-14x trailing earnings - priced as a dying industry while converting a third to nearly half of revenue into free cash [28]. These two allocations cannot both be right. “Right in total, wrong in location” is therefore a coherent resolution of the $17 trillion residual: builder capitalization compressing toward replacement cost while rider capitalization expands. Relocation redistributes the requirement without relaxing it - whoever earns the rents, the external revenue must still arrive at 3-6.5 percent of world GDP for the aggregate pricing to hold. Section 10 examines the first violent market experiment on exactly this question.

5. Historical Calibration: IT Spending and GDP

Gartner's April 2026 revision puts worldwide IT spending at $6.31 trillion for 2026, up 13.5 percent - about 5.3 percent of world GDP (~$120T). Hardware is $788 billion of datacenter systems plus $856 billion of devices, about $1.64 trillion or 1.4 percent of world GDP; software runs $1.44 trillion, IT services $1.87 trillion, communications services $1.36 trillion [2].

Segment	2025 ($B)	2025 growth	2026 ($B)	2026 growth
Data center systems	489	+46.8%	788	+55.8%
Devices	783	+8.4%	856	+8.2%
Software	1,244	+11.9%	1,440	+15.1%
IT services	1,719	+6.5%	1,870	+8.7%
Communications services	1,304	+3.8%	1,360	+4.8%
Total	5,540	+10.0%	6,310	+13.5%

Table 5. Worldwide IT spending by segment. 2025 figures and 2025 growth from Gartner's October 2025 release [3]; 2026 from the April 2026 revision [2] (2026 devices growth as reported against revised 2025 base).

The historical shape is more interesting than a simple rise, and it disciplines the Section 4 requirement. The global ratio has essentially round-tripped: roughly 5.3 percent of world GDP around 2008, drifting down to about 4.3 percent by 2019-20 as communications stagnated in absolute terms while GDP grew, then inflecting back - 4.8 percent in 2025, 5.3 percent in 2026 (historical shares are authors' estimates from Gartner series and IMF GDP data). The headline share sat in a 4.3-5.3 percent band for two decades, and 2026 is the first decisive break above the band's ceiling. Composition rotated completely underneath: telecom fell from nearly half the total to about a fifth; software went from roughly $300 billion in 2010 to $1.44 trillion; and datacenter systems - the line that matters for this report - sat at $150-200 billion for essentially the entire 2010s, four to five percent of IT spending, before going roughly $243B (2023) to $333B (2024) to $489B (2025) to $788B (2026): a tripling in three years, now 12.5 percent of all IT.

The mechanism behind two decades of flatness is worth naming: compute consumption grew exponentially throughout, but hardware price deflation - Moore's law plus competitive supply - absorbed it, holding the nominal line flat. The current breakout is partly the suspension of that deflation: accelerators priced at 75 percent gross margins and HBM at doubled prices mean the datacenter line now grows with volume and price together, and Gartner's own commentary attributes its upward revisions substantially to memory pricing [2]. Some fraction of the IT share of GDP finally rising is the scarcity-rent regime showing up in category data. As a base rate for Section 4, this is sobering: the share needed forty years to reach five percent and then did not grow for twenty, and the valuation math needs a second IT-sector-sized wedge - 3-6.5 percent of world GDP - inside a decade. The one measurement caveat cutting the other way: Gartner's perimeter excludes the ad-funded consumer internet (Google's and Meta's roughly $700 billion of revenue is not counted as IT spending), and the BEA's broader digital-economy account puts the fuller US tech share near ten percent of GDP, up a couple of points over two decades. The economy can grow a multi-point tech wedge; it did so once, over about twenty-five years, funded largely by advertising's fixed share of GDP. The open question is whether labor substitution funds the second one faster.

6. The Buildout Precedents: Railways, Railroads, Telecom

Three completed buildouts frame the AI cycle: Britain's Railway Mania of the 1840s, the American railroad expansion from the Civil War through the 1890s, and the telecom buildout of 1996-2002. Each was, in its day, the largest private-capital infrastructure program in history; each transformed its economy; each treated its investors badly in a characteristic way. Table 6 compares them; the records follow.

6.1 Three records

Britain, 1844-1852. Railway investment peaked near five to seven percent of GDP in 1847 - a share no private program has matched since [30]. The mania's texture matters more than its size. Dividends rose from 4.4 percent in 1843 to 7 percent in 1847 while consols paid 3, and share prices rose 106 percent into 1845 - then both collapsed together, dividends to 2.4 percent by 1852 and prices below their starting value [30]. Campbell's reconstruction yields the finding this report leans on in Section 11: railway shares were not obviously mispriced even at the peak, because prices tracked dividends up and down; the catastrophe was aggregate, not per-security. The expansion, moreover, was carried out mostly by established, hitherto-profitable railways through rights issues - respectable incumbents, not fly-by-night promotions - and the new lines were simply, collectively, unremunerative [30]. Britain received a national network it would never have voted to fund through taxation, and railway shareholders spent decades earning less than the government's own perpetuities paid.

The United States, 1865-1900. Between 1865 and 1875 the country roughly doubled its rail network - 35,085 miles to 74,096, with construction peaking at 7,439 miles laid in 1872 [34] - and railroad capital reached roughly $4.5 billion when the entire banking system's capital was $720 million and the federal debt $2.3 billion [28, 29]. In January 1870, 87 percent of shares traded on the New York Stock Exchange were railroad shares, and from 1870-74 roughly seventy percent of all railroad securities issued in London were American - foreign capital drawn by 6.5 percent gold bonds against far lower consol yields [28]. The financier's failure was the underwriter's: Jay Cooke contracted to place $100 million of Northern Pacific bonds, sold less than $20 million, and ended up owning three-quarters of the railroad he was financing when his house failed on September 18, 1873, closing the NYSE for ten days - the first closure in its history [28, 29]. By 1876, 134 railroads were in default on $500 million of roughly $2 billion outstanding; by January 1877 more than 18 percent of national mileage was in receivership; and the 1893 panic repeated the cycle at larger scale, with a fifth of mileage - Union Pacific, Northern Pacific, Santa Fe, and Reading among them - in receivers' hands at the March 1894 peak, a quarter of aggregate capitalization passing through receivership across 1893-97 [29, 34]. The boom-decade peak came later: construction quadrupled from 2,665 miles in 1878 to 11,569 in 1882 [34], railroads absorbed an estimated fifteen percent of all American capital formation across the 1880s - roughly six percent of GDP at peak - and 1890s capitalization carried the forty-percent water of Section 3.4 [29]. And where roads survived, competition took the returns: revenue per ton-mile fell from roughly 2.2 cents in 1870 (Fishlow's estimate - no all-road series reaches back that far) to 0.73 cents in 1900, both nominal [29, 34], and rate wars drove the New York-Chicago through rate from $1.88 toward twenty cents [28]. Every additional mile made the network more valuable to the country and less valuable to the men who had paid for it.

Telecom, 1996-2002. Between the Telecommunications Act and 2002 the sector raised on the order of $1.5-2 trillion in equity and debt and roughly tripled annual capex, peaking around $110-120 billion per year in the US in 2000 before collapsing by half within three years. The demand forecast underneath - internet traffic doubling every hundred days - was off by roughly an order of magnitude (actual doubling was close to annual; Odlyzko spent years dismantling the meme [18]), while DWDM multiplied capacity per fiber strand roughly a hundredfold mid-buildout. By 2002 an estimated 3-5 percent of installed long-haul fiber was lit and bandwidth prices on major routes had fallen more than ninety percent. WorldCom - $107 billion of stated assets, $11 billion of fraudulent accounting - became the largest bankruptcy in US history to that date; Global Crossing put roughly $10 billion of audited capex into its network and sold control out of bankruptcy for $250 million, a reorganization value near one cent on the dollar of peak book assets; Level 3 survived a 99 percent drawdown. European carriers paid about EUR 110 billion for 3G spectrum in 2000, mostly written off - a direct shareholder-to-treasury transfer. FCC Chairman Michael Powell told the Senate in July 2002 that roughly $2 trillion of market value and nearly $1 trillion of debt had been lost; The Economist's leader the same month put sector debt at about $1 trillion, up to half of which might never be repaid [21], and sector-level studies through the 2000s and 2010s consistently put telecom returns on invested capital at 5-8 percent against 7-9 percent costs of capital. Persistently, on both sides of the Atlantic, the industry operating the internet's transport layer earned less than its capital cost.

	UK Railway Mania (1840s)	**US Railroads (1865-1900)**	**Telecom (1996-2002)**	**AI (2023-)**
Peak investment	~5-7% of GDP (1847)	~6% of GDP (1880s peak); ~15% of capital formation across the decade	~1.0% of US GDP (2000); more than $500B in five years	~1.2% of US GDP (2025); ~$725B big-four guidance for 2026
Financing	Partly-paid equity and calls; incumbents' rights issues	Bonds, ~70% sold abroad (London, 1870-74); federal land grants; promoter construction companies (Credit Mobilier)	Public equity and high-yield debt; vendor receivables financing (Lucent, Nortel); state 3G fees	Operating cash flow, then debt ($108B in 2025, ~$175B 2026E), SPVs (Hyperion $27.3B), equity-linked vendor deals
Dominant asset life	30-50 yr way and works	30-50 yr roadbed; shorter rolling stock	~25 yr fiber; shorter electronics	~5 yr IT equipment (62% of stock); 15-30 yr shells and power (38%)
Private return	Dividends 7% (1847) to 2.4% (1852); long-run below consols	20% of mileage in receivership by 1877, ~25% again by 1894; ton-mile revenue -61%, 1870-1900	ROIC 5-8% vs 7-9% WACC; ~$2T equity and ~$1T debt erased; capex halved 2000-03	Open; realized rents ~$330-350B/yr, mostly supply-chain scarcity; sector Q 15-23x
Toll-booth winners	W.H. Smith bookstalls; Thomas Cook excursions	Adams Express, American Express, Pullman - asset-light riders on others' track	Google, Amazon, Netflix; content platforms funding ~4 of 5 new transatlantic-cable dollars by 2018-19	Contested - incumbent software vs model layer vs users; July 2026 the first live test (Section 10)
Loss absorption	Shareholders via calls; diffuse	Foreign bondholders (~$600M, 1873-79); domestic equity; receivership courts	Diffuse equity (pensions, retail); mild 2001 recession	Index public via cap weight; credit if the debt migration continues; ratepayers at the margin

Table 6. Four network buildouts compared. Sources: [18]-[21], [28]-[31]; AI column, this report. Express-company and per-name figures relayed via [28]; see Data Provenance.

6.2 The productive-bubble thesis

From a global rather than shareholder perspective, all three records admit a stronger reading: the networks were built with investor money that never saw a return - covertly socialized infrastructure programs, neither consensual nor centrally planned, but effective. Nordhaus's estimate is the clean theorem behind it: even in successful innovation, producers capture on the order of 2.2 percent of the social surplus they create [19]. A productive bubble is the degenerate case where private capture goes negative - investors pre-fund infrastructure whose surplus they cannot appropriate, and the crash is the moment that fact is marked to market. This is Perez's installation/deployment structure and Janeway's productive bubble [20]: the crash transfers assets to new owners at written-down cost basis, and that basis is what makes deployment-era services cheap. Google bought dark fiber from bankrupt carriers for pennies mid-decade; YouTube launched in 2005 into a bandwidth regime ninety percent cheaper than any rationally priced buildout would have produced. Because the losses were equity-financed and diffuse - pensions and retail absorbed them - the 2001 recession was mild; bank-funded buildout busts (1873, 2008) propagate instead of dissipating.

Two complications keep the frame honest. First, within the investor class it was a transfer, not a shared sacrifice: insiders at Qwest and Global Crossing sold billions near the top, conflicted research and the IPO machinery pulled late retail money in, $11 billion of WorldCom's numbers were fabricated - and the Mania had George Hudson paying dividends out of capital. Consensual socialization it was not. Second, Odlyzko's sharper point: the bubble funded the fundable layer, not the binding one. Long-haul transport was a small fraction of total network cost with easy entry, so speculative capital pooled there; the actual bottleneck - the last mile - was built later, slowly, by incumbents on regulated economics. Bubbles allocate by legibility and venture-shape, not by bottleneck. The AI equivalent of the last mile is plausibly power interconnection and enterprise integration, which is roughly where capital moves slowest today.

6.3 Asymmetries and the deflation clock

One asymmetry deserves heavy weight in any mapping to AI: the gift mechanism requires the capital stock to outlive the crash. Fiber in conduit had twenty-five-year life and near-zero carrying cost - it could sit dark waiting for YouTube; rail roadbed lasted half a century. The AI stock modeled in Section 3 is 62 percent IT equipment on roughly five-year lives: of the four episodes, this is the weakest bequest case. An AI overbuild does not bequeath, it evaporates, leaving the shells and power infrastructure (roughly 38 percent of the stock), the trained workforce, and whatever persists in open weights. The ton-mile series doubles as the cycle's clock: railroad ton-mile revenue deflated by roughly two-thirds in nominal terms over thirty years, while AI inference prices routinely fall as much in a single year. The commodity-deflation phase that consumed a generation compresses here into quarters - the network's output becomes cheap before its owners have depreciated the plant, which is the five-year life restated from the price side. And the financing drift cuts the other way: dot-com losses were equity-diffused, while this cycle's debt migration, private credit, and special-purpose vehicles move it toward the banked-bubble category, where socialization stops being benign and becomes literal - bailouts and ratepayers, the latter channel already active in utility bills.

7. The Second Harvest: What the Losses Actually Bought

Take the socialized-buildout claim at full strength and the retrospective justification of the telecom writeoff was never merely cheap video. The overbuilt network did two things no planner ordered. It induced humanity to transcribe itself - the web, the forums, the code repositories, trillions of tokens produced as exhaust by people using bandwidth priced below its buildout cost - and it grew the ad-funded engine that later paid for the laboratories. The corpus and the cash flows were both second harvests of the glut. On the longest accounting, the 2000-02 losses were the capital contribution to machine intelligence: privately negative, socially incalculable, with the payoff arriving twenty years late in a form nobody was pricing. That is not a metaphor stretched over the facts; it is the causal chain.

Open weights are then the piece that makes this cycle's version of the bequest structurally stronger than fiber. Fiber persisted because glass in conduit does not rot. Weights persist because of irreversibility plus zero-cost reproduction - and they add a property no prior infrastructure had: the artifact teaches its own manufacture. Every served token is a lesson; distillation means the product can be interrogated into yielding its replacement. Nordhaus found producers capture about two percent of innovation surplus on average; for this good that starts to look like a ceiling, because appropriability leaks through the product's own function. The physical form is maximally unpossessable - the crown jewels of a multi-trillion-dollar complex are a few hundred gigabytes, which fit in a backpack and therefore in no vault. Export controls govern chips because nobody can durably govern files. Two refinements keep the analogy honest. The dark fiber of this cycle is not the weights already published - those are lit and carrying traffic - but the elicitation overhang: capability resident in released artifacts that current scaffolding has not yet extracted. And the crash scenario is synergistic for the commons rather than destructive: a bust means fire-sale accelerators the way 2002 meant two-cent bandwidth, and written-down compute plus free weights is exactly the deployment-era cheapness the next platform gets built on. The rent capitalization of Section 3 is crash-exposed; the endowment is crash-hedged.

The friction is real and should be stated against the grain of the optimism: the gift is parasitic on the race. Open releases are strategic byproducts - complement commoditization, positioning, recruiting - funded by the very rent expectations sized above. The floor rises only while the frontier is contested; the day the funding stops, the last release freezes as the permanent public endowment, trailing closed capability by an unknowable gap. And diffusion through this commons selects on capability per dollar, not on wisdom - a discrimination problem with no filter at all, because the unpossessable is also the unrecallable. There is an old resonance here that requires no forcing: the provision that spoiled when hoarded, save the portion set aside; closed frontier value decays on roughly that schedule - months to obsolescence - while the released portion keeps. Property was the wrong category from the start; the balance sheets are merely the last to find out.

8. The Circular Economy: The Ouroboros Formalized

A recent Atlantic essay describes the AI economy as a trillion-dollar ouroboros of buying and selling, investment and equity staking, largely among a handful of firms - Big Tech advancing money to AI startups to buy cloud services from Big Tech [10]. Two underlying sources supply the microstructure for precisely the aggregate claims built above: the funding gap and the margin-stacking constraint.

8.1 The firm-level instance: OpenAI

PitchBook's analysis (Harrison Rolfes) is the firm-level version of the Section 3 inversion, and it closes consistently: to justify an $852 billion valuation, OpenAI must generate $95-105 billion of free cash flow by 2030 - roughly the same 8-12x forward-rent multiple the aggregate implied - while on Q1 trajectory it loses $10-30 billion that year instead [5]. (Free cash flow: operating cash flow minus capital expenditure - the cash actually left after running and investing in the business; FCF margin is that as a share of revenue.) Q1 2026 revenue was $5.7 billion at a negative 122 percent adjusted operating margin: $2.22 spent per dollar earned. Even OpenAI's own plan - roughly $280 billion of 2030 revenue, up from $13.1 billion in 2025, with its infrastructure target cut from $1.4 trillion to about $600 billion [11] - requires a ~36 percent FCF margin at scale, better than Microsoft's or Apple's mature cash economics, achieved within four years while buying compute at scale against Google, Anthropic, and free open weights. The infrastructure target itself is mostly not owned assets but long-term purchase commitments: named pledges of roughly $250B to Azure, $300B to Oracle, $138B to AWS, $22.4B to CoreWeave, $10B to Broadcom, up to $100B from Nvidia (still a letter of intent per Nvidia's CFO as of December 2025), plus a 6GW AMD deal carrying warrants for up to ten percent of that supplier [7, 8, 9]. Roughly $820 billion of commitments stand against a $25-33 billion current revenue run rate - and this is the designated demand terminus of the whole loop, the node where external revenue is supposed to enter, with 85 percent of its 800-900 million weekly users on free tiers [5, 6]. The February 2026 markdown of the target from $1.4T to $600B was therefore not housekeeping:

the terminus wrote down its own pledged draw by roughly $800 billion, and those pledges are exactly the flows other layers had capitalized.

8.2 Consolidation, double-capitalization, and margin stacking

Draw the accounting perimeter around the complex and eliminate intra-group transactions, as consolidation would for any conglomerate: every circular deal vanishes in elimination entries, and what remains is the external P&L - roughly $300-400 billion of outside revenue against $700 billion to $1 trillion of cash out. That gap is the socialization flow, running now, continuously - the structural difference from telecom, which socialized at the writeoff. The funding legs are index flows (the largest seven technology companies' ~35 percent S&P 500 weight makes every passive retirement dollar a mechanical contribution), the debt migration, and sovereign capital. The Atlantic's what-happens-to-one-happens-to-all is then simply the observation that intra-complex claims net to zero: the complex is economically one consolidated entity whose equity tranches happen to trade separately.

The sharpest structural feature is double-capitalization of a single uncertain flow. The same expected OpenAI spending appears as Oracle's remaining performance obligations - RPO, the accounting disclosure of contracted-but-undelivered revenue, which exploded to roughly half a trillion dollars largely on that one contract and repriced Oracle's equity enormously - as Nvidia's backlog, as CoreWeave's contracted revenue, and as hyperscaler committed-spend disclosures. Each layer capitalizes its margin slice of the flow into its own market capitalization, while OpenAI's $852 billion capitalizes the end revenue that must fund all of them. A trillion dollars of commitments is not a trillion of demand; it is one ~$280-billion-per-year aspiration pledged severally, which is why tail correlation inside the perimeter approaches one. The flagship toll-booth exhibit shows the same contamination directly: Microsoft's commercial remaining performance obligation reached $678 billion in mid-2026, up 84 percent year over year - widely cited as the railroad-freight-contract evidence of durable enterprise demand - and includes OpenAI's $250 billion Azure commitment [26, 28]. Railroad freight contracts were signed by shippers with cargo; a third or more of this one is signed by the loop's own negative-margin terminus. And the margin-stacking constraint is already binding: the April renegotiation capping Microsoft's revenue share at $38 billion through 2030 - without which, per PitchBook, positive free cash flow is not possible - is rent reallocation inside the loop, by private negotiation, before external revenue has arrived [5]. The layers' summed margin assumptions exceeded what the flow could fund, and the reconciliation has quietly begun.

8.3 Equity-linked vendor financing and the corrupted signal

The instrument set rhymes with 1999 but differs in one load-bearing way. Lucent and Nortel did vendor financing through receivables, concentrating customer credit risk on their own working capital. This cycle's version is equity-linked: supplier investments in customers, warrants granted to a customer, utilization backstops (Nvidia's arrangement to absorb unused CoreWeave capacity), compute credits structured as investment [7, 8, 9]. Equity-linked financing is system-safer - losses diffuse to shareholders rather than triggering a receivables death spiral - but epistemically worse, because it contaminates the demand signal itself. When the supplier funds the purchase and books it at 75 percent gross margin, and when Google's and Amazon's reported earnings include paper gains on their laboratory equity stakes (a point Fortune quantified in April 2026), the market's instruments for measuring whether external demand exists are themselves inside the loop. The contamination is now quarterly routine: Microsoft's July 2026 print included a $3.2 billion gain on its Anthropic stake - roughly nine percent of the quarter's net income marked from lab equity inside the loop [26]. Price discovery degrades because nearly every transaction carries a financing leg. Add OpenAI's DeployCo joint venture guaranteeing private-equity investors a 17.5 percent annual return [9] - equity risk repackaged as a debt-like claim - and the marginal capital is no longer risk-bearing in Perez's sense at all; it is rented, which is how installation phases end.

The banked-bubble drift now has concrete instruments. Meta's Hyperion campus in Louisiana is financed through a special-purpose vehicle in which Blue Owl holds 80 percent: $27.3 billion of senior secured notes at a 6.581

percent coupon maturing 2049, secured not by the datacenter but by Meta's promise to pay rent from 2029 plus a residual-value guarantee - $27 billion of capacity debt held off balance sheet against a lease [28]. And credit markets already tier the loop that equity trades as one theme: Meta's SPV paper at 6.6 percent against CoreWeave's term loans at 11-15 percent and a 12.3 percent weighted short-term borrowing rate - a spread of five to six hundred basis points inside the same buildout, the tail correlation priced by lenders and ignored by index weights [28]. The IPO wave - OpenAI and Anthropic listing alongside Databricks [5] - is the 1999 rhyme: exposure migrating from strategic balance sheets to the index-holding public precisely as the internal reconciliation begins.

8.4 The ancestor: Credit Mobilier

The related-party graph has an exact ancestor, one buildout older than Lucent. Credit Mobilier of America was the construction company owned by the Union Pacific's own promoters, which billed the railroad on the order of $94 million for construction costing perhaps $50 million and distributed discounted shares to congressmen for protection; the scandal broke in 1872 and the censures came in 1873, the year the boom itself did [32]. The structure's essence: the profits of the buildout were booked by insiders on the construction leg while the operating asset was born overloaded with obligations - construction-phase rent extraction wearing the costume of network demand, the corrupted signal a century and a half before vendor financing. Every generation's version differs in instrument and rhymes in function: Credit Mobilier's construction invoices, Lucent's receivables, Nvidia's equity stakes and warrants. The diagnostic question is constant across all three: who profits if the network is built, regardless of whether it is ever profitably run?

9. Vendor Financing, 1996-2002: Lucent, Nortel, Winstar

The precedent deserves its detail, because both protagonists were blue-chip institutions. Lucent - Bell Labs plus Western Electric, spun out of AT&T in 1996 - was briefly the most widely held stock in America, peaking near $250 billion of market value on about $38 billion of revenue. Nortel traced to 1895 and at its July 2000 peak represented roughly a third of the entire Toronto index at about C$400 billion. Both lost ~99 percent of their value within two years; Lucent merged into Alcatel in 2006 for $13.4 billion - a nineteenfold markdown from peak - and Nortel ceased to exist.

The mechanism: the 1996 Act spawned hundreds of competitive local exchange carriers - thinly capitalized startups whose business plan was borrowing money to buy network equipment - and the vendors competed for those orders by supplying the purchase money themselves. Lucent's vendor-financing commitments reached roughly $8 billion, Nortel's several billion more, the industry perhaps beyond $20 billion. The accounting made it lethal: a vendor-financed sale books as revenue immediately, at full margin, with a receivable on the asset side - so the income statement showed booming demand while the balance sheet quietly accumulated credit exposure to customers solvent only while capital markets stayed open. Demand was partially synthetic - the vendors' own capital returning as orders - and the street extrapolated the revenue growth without consolidating the loop. When CLEC funding closed in 2000-01, customers died, receivables became writeoffs, and the extrapolated demand evaporated simultaneously. The canonical case is Winstar: Lucent committed $2 billion of financing to a fixed-wireless CLEC; Winstar went bankrupt in April 2001 and sued Lucent for $10 billion for cutting off the credit line - the customer suing the supplier for refusing to keep funding its own purchases, the loop stated as a legal claim.

Each company added its own pathology. Lucent pulled demand forward with quarter-end discounts and distributor stuffing, restated $679 million of fiscal-2000 revenue, and ultimately faced SEC charges over roughly $1.1 billion of improperly recognized revenue [22]; the stock went from $84 to under a dollar, survival required spinning off Avaya and Agere and cutting headcount from ~130,000 toward ~35,000. The signature is the triple hit - revenue restated, receivables written off, credibility gone - each amplifying the others. Nortel's variant was paying about

$32 billion in inflated stock for acquisitions and writing essentially all of it off - a $19.4 billion single-quarter loss in mid-2001, roughly $27 billion for the year - followed by a second act in which post-crash management released accounting reserves to manufacture a 2003 return to profitability, triggering executive bonuses, terminations for cause, two multi-year restatements, and criminal trials ending in acquittal in 2012 but never in recovered credibility. It filed in January 2009 - Canada's largest bankruptcy [23]. The estate sale connects to the residue thread: the operating businesses fetched a few billion in pieces while the single most valuable asset was the patent portfolio, sold for $4.5 billion to an Apple-Microsoft-Ericsson consortium that outbid Google, whose opening bid was literally $3.14159 billion - pi. The durable product of a century-old institution was pure intellectual property, purchased for exclusion. Cross-border litigation over splitting the estate burned roughly $2 billion in professional fees over eight years, and some twenty thousand pensioners took devastating losses.

What the detail adds to the AI mapping. First, the vendors acted knowingly, inside a prisoner's dilemma: if Lucent would not finance Winstar, Nortel or Cisco would, and land-grab logic made unilateral prudence equivalent to ceding share. The same structure operates now - each equity-linked deal defensible as positioning, the ensemble a machine for manufacturing the demand signal. Second, the failure mode differs by instrument: receivables financing put customer credit risk on the vendor's own working capital, so death came fast and triple-barreled; equity-linked losses mark to a line investors already treat as speculative, so an unwind would be slower and less mechanically self-reinforcing - genuinely safer plumbing around the same corrupted signal. Third, collateral: repossessed CLEC networks were worth nearly nothing to anyone else, whereas repossessed GPUs are fungible - though fungible into a glut, which is how two-cent bandwidth happened. Fourth, any unwind's complexity: Nortel's estate had one jurisdictional web to untangle and consumed $2 billion in fees; the OpenAI-Microsoft-Oracle-Nvidia-SoftBank claim graph would make that fight look like small claims. The cheerful coda stands: Winstar's stranded assets and the CLECs' dark fiber became the cheap substrate of the deployment era. The vendors financed a commons and were repaid in bankruptcy claims.

10. The July 2026 Rotation: Rent Location Goes Live

The rent-location question stopped being theoretical in July 2026, but the experiment began nine months earlier. From October 2025 the market ran the builder-versus-rider trade at index scale, in the builders' favor: enterprise software sold off on fears that agents would cannibalize seat-based spending - the debate acquired a name, the “SaaSpocalypse” - with Salesforce down 33 percent and Adobe down 31 percent for the year by late July, while the buildout's second derivative ran [27]. Situational Awareness LP - founded in mid-2024 by Leopold Aschenbrenner, a former OpenAI researcher, on $225 million from Patrick and John Collison, Nat Friedman, and Daniel Gross - was that trade in its purest form: long memory, fuel cells, neoclouds, and bitcoin miners converting substations into compute; short the application incumbents; roughly four times levered [24, 28]. It returned 439 percent net in the first half of 2026 and reached about $45 billion of assets with a staff of roughly a dozen, and its July 24 letter to investors invited additional capital [24, 25].

Twenty sessions inverted both legs. The fund's disclosed longs fell 36-55 percent in July - Sandisk -55, Nebius -46, Bloom Energy -46, CoreWeave -39, Micron -36 - while the software side surged: for the full month, Workday +31, Adobe +22, Intuit +22, Salesforce +18, and Microsoft +25, against a Nasdaq down roughly ten percent and an S&P down two, with Nvidia about flat [27, 28]. This was not an AI crash; the index barely moved. It was a violent migration out of the leveraged, capital-hungry end of the complex into the profitable, asset-light end - out of the builders, into the toll booths. Margin calls from Bank of America, Goldman Sachs, and JPMorgan forced the sale of the entire public book, longs and shorts together, roughly $16 billion, to Citadel in a single block at about a ten percent discount before the July 30 open [24]. Assets finished near $10 billion - roughly half of it a retained private stake in Anthropic, which a spokesman said was not for sale - with the fund still up about 80 percent for the year and reopened to subscriptions on August 1 [24].

The tape's testimony must be read with care, because it cuts both ways. Every core long bottomed on July 29 - margin-call day - and snapped back violently the moment the block cleared: both sides of the book moved in the fund's favor the day the book stopped being the fund's, which is the cleanest available statement that July's prices measured positioning, not truth [25]. The Microsoft print is the controlled experiment inside the event. Into its July 29 report the stock carried its highest short interest in a decade - 92 million shares, the largest increase among the seven largest technology companies, with no covering into earnings - and a 19 percent year-to-date decline: the market had spent 2026 treating Microsoft as a doomed capex name [26]. It reported Azure accelerating to 43 percent growth - the fastest in four years, past $100 billion for the fiscal year - and held rather than raised its capital-spending guidance, and the stock rose about 16 percent, adding $480 billion in a single session, as the shorts covered and the market's largest forced seller exited the same morning [26]. Forced to choose between Microsoft-the-builder and Microsoft-the-rails, the market chose the rails; the violence of the choosing was flow. The choice then held: through the turn of the month the rotation extended rather than reversed [27]. And inside the celebrated print sat Section 8's contamination, live: a $3.2 billion gain on the company's Anthropic stake - roughly nine percent of the quarter's net income marked from lab equity inside the loop - alongside the company's first-ever voluntary retirement program in the same quarter it passed thirty million paid Copilot seats [26].

Three lessons for the framework, none of them the consensus one. First, the event is the first mass repricing from the market's rent allocation toward history's - builders toward replacement cost, riders toward rent capture - and it occurred without any aggregate crash, exactly as the location corollary permits: the level and the location of the rents are separate questions, and July moved only the second. Second, the blowup does not adjudicate the decade thesis. Early and levered is punished identically to wrong; a fund can be carried out of positions that later prove correct, and the public post-mortems split on precisely this point within days [25, 28]. The residual irony is instructive without being conclusive: when the second-derivative book burned, the asset that survived was equity in a model laboratory. Third, the toll-booth reading carries two qualifications its promoters omit. Seat-priced software is a derivative claim on headcount: the express companies won because package traffic exploded, while a seat-based toll booth sits on a road whose traffic is the thing being automated - and Microsoft ran its first voluntary retirement program in the same quarter it sold thirty million Copilot seats. Survival requires repricing from seats to work faster than usage-native entrants undercut, and the incumbents' agent products - consumption-priced, growing at triple-digit rates from small bases - are that repricing attempted in real time. And the historical toll booths did die: parcel post in 1913, consolidation into American Railway Express in 1918, Pullman by consent decree - killed by the state, not by competition [31]. Moats built of standards, audit trails, and validated records are political artifacts, and regulatory tail risk is what concentration accumulates. A provenance caution completes the record: the sharpest public post-mortem of the episode doubles as marketing for toll-booth stock recommendations [28]. Its history checks out; its certainty is merchandise.

11. Synthesis and Watch Variables

The through-line of this report is a single decomposition applied at every scale. Market value equals replacement-cost assets plus capitalized rents; the assets are ~$1.2 trillion and the capitalized rents ~$17 trillion, so the entire live dispute concerns a rent stream that must grow several-fold and persist. The rents currently earned are predominantly scarcity rents in the supply chain - the mean-reverting kind - and they are simultaneously embedded in the asset book values, so competitive normalization shrinks numerator and denominator together. The revenue that must eventually fund the rents has no sufficient source except labor substitution, at a GDP share that history says takes decades to grow, and the interim is bridged by a circular financing structure that consolidates to a negative external cash flow funded continuously by index capital, debt, and sovereigns - a socialized buildout in real time rather than at the crash. The precedents say the private outcome and the social outcome can diverge completely: three networks were built by capital that was never repaid, and society received the substrate of the modern world - including, on the longest accounting, the corpus and the cash engine that produced machine intelligence. History adds that the profits, where they existed, went to asset-light riders on the network rather than its owners; July 2026 was the first mass repricing toward that allocation. Whether this cycle's version bequeaths or evaporates turns on asset lives and on what fraction of capability persists in open, unpossessable form.

The strongest form of the whole argument comes from the railway literature and deserves the closing word on method. Campbell's reconstruction of the 1840s found that railway shares were not obviously mispriced even at the peak - prices rose 106 percent and then fell below their starting value, but dividends rose and fell along the same path - even though the Mania's investors took large losses and the network build was, in aggregate, ruinous [30]. A sector can overbuild catastrophically without any individual security being mispriced against its own expected cash flows. The $8-17 trillion residual of Section 3 therefore requires no claim that anyone is irrational about Nvidia or Microsoft; fallacy of composition suffices - each layer fairly priced against margin assumptions that cannot jointly be satisfied by the end flow. That is a harder claim to dismiss than “the market is wrong,” and it is the claim this report intends.

The variables worth watching, each tied to a specific failure or falsification mode developed above: the CoWoS supply-demand gap closing through 2026 and memory fabs landing 2027-28 (scarcity-rent normalization; the bullwhip); external AI revenue against the ~$1.7 trillion breakeven floor and the 3-6.5 percent-of-GDP ceiling (the transition from capex-funded to demand-funded rents); cloud AI growth rates sustaining above ~50 percent (the falsifier of the pessimistic case); the seat-to-work repricing of application software and whether the July 2026 rotation holds (the live rent-location experiment); the credit spread between the complex's investment-grade core and its levered edge (the lenders' pricing of the tail); the debt and private-credit share of buildout funding (equity-diffused versus banked-bubble loss distribution); further intra-loop rent renegotiations of the Microsoft-cap type (margin stacking made visible); RPO and backlog concentration against single counterparties (double-capitalization exposure); the IPO absorption of OpenAI, Anthropic, and Databricks paper (the migration of exposure to the public); and the cadence and capability gap of open-weight releases (the size of the permanent endowment if the funding stops).

Data Provenance

Figures in this report divide into three classes. **Reported:** company disclosures and named third-party research - hyperscaler capex and guidance, Nvidia and TSMC results, Gartner spending series, PitchBook's OpenAI analysis, the CNBC/WSJ/FT accounts of the Situational Awareness unwind, and Microsoft's FQ4 2026 results. **Estimated by the authors:** the AI-attribution shares of capex, the vintage capital-stock model and all of Tables 1-4, the competitive-cost repricing, current external AI revenue ($300-400B), realized rent composition (~$330-350B/yr), historical IT-share-of-GDP ratios, and the labor-decomposition arithmetic. **Historical literature:** the railway, railroad, telecom, and vendor-financing records, cited below. An August 2026 primary-verification pass re-anchored the depreciation-extension inventory and Amazon's reversal to the companies' own filings [33] (correcting the mixed operating/net series relayed by [28]), the mileage and ton-mile series to Historical Statistics, NBER Macrohistory, and Fishlow [34] (restating the 1870 ton-mile endpoint as Fishlow's ~2.2-cent estimate), and the Moody's and Bain projections to their publishers [35, 36]. Per-name July 2026 long/short returns, the nineteenth-century express-company financials, the Cooke and NYSE-share detail, and the loop's private financing terms are still taken from [28] - a marketing document - and have not been independently verified line by line; the aggregate account of the fund unwind is verified against [24]-[25], the Microsoft print against [26], the software-rotation aggregates against [27], and the railroad statistics against [29]-[30] and [34] where sources exist. Where a range is given, the range is the claim. Nothing here is investment advice; it is an accounting framework with explicit assumptions, built to be argued with.

Appendix A. Capital Stock Model

Vintage gross capex as in Table 1. Asset mix and straight-line lives: IT equipment (accelerators, servers, networking, memory, storage) 62 percent at 5 years; power and cooling infrastructure 23 percent at 15 years; shell and land 15 percent at 30 years; mid-year convention per vintage. The five-year IT life is corroborated by Amazon's January 2025 reversion of a subset of server lives from six years to five, disclosed as a response to the pace of AI hardware development [26, 28]. Add-ons at mid-2026, net (low/central/high, $B): upstream fab plant AI-allocated 105/125/145; power and grid 45/65/85; model IP at cost excluding compute 35/55/75. Competitive-cost adjustment: accelerator and networking spend taken as 75 percent of the IT stock, repriced to 45 percent of paid price (competitive-margin replacement); other IT repriced to 75 percent; infrastructure and shell at cost. Rent flows in Table 3 use required flow = rent capitalization times (r - g) for the growing case (r = 8%, g = 3%), times r for the flat case, and times (r + d) with d = 10% for the decaying case. Sensitivity: moving every vintage to its high case and Nvidia's implied IT share to 55 percent raises the central net stock by roughly $200B; the rent capitalization changes by less than 2 percent, because it is dominated by the market-value term.

Appendix B. Revenue Requirement Model

Required after-tax profit = required rent flow (Table 3, $18T attribution) + 8 percent normal return on a projected 2030 net stock of ~$5T (from ~$5.3T big-four capex 2025-2030 plus the rest of the ecosystem, net of depreciation). External revenue = profit / consolidated net margin, shown at 12, 20, and 30 percent. GDP bases: world ~$135T and US ~$35T nominal in 2030 (~5 percent nominal growth from ~$118T and ~$30.5T in 2026, IMF-consistent). Breakeven floor = depreciation on the 2030 stock (~$940B/yr at the Appendix A mix, net-to-gross ~0.77) + operations (~$275B) + tax-adjusted cost of capital (~$480B) = ~$1.70T/yr = 1.3 percent of 2030 world GDP. Labor decomposition: revenue R = capture share times labor cost addressed; at 30 percent capture, R of $4.0/5.5/7.0T implies $13.3/18.3/23.3T addressed = 19/26/33 percent of global labor income (53 percent of world GDP).

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[8] E. Forgash and A. Ghosh, Bloomberg, "OpenAI, Nvidia Fuel $1 Trillion AI Market With Web of Circular Deals" (October 7, 2025); N. Smith, "Should we worry about AI's circular deals?" Noahpinion (October 2025), noahpinion.blog/p/should-we-worry-about-ais-circular; S. Rasgon, Bernstein Research note (September 2025), as reported.

[9] AI Circular Economy tracker (ai-circular-economy.com, 2026): compiled deal terms including OpenAI compute commitments by counterparty, Nvidia CFO letter-of-intent status (December 2025), DeployCo return guarantee (May 2026), and the Fortune (April 2026) analysis of paper gains on laboratory equity in Google and Amazon earnings. Secondary compilation; individual deal terms as reported by Bloomberg, FT, and WSJ.

[10] The Atlantic (July 2026), essay on AI valuations and circular investment, as excerpted by the requester; the ouroboros characterization and the PitchBook FCF citation appear there.

[11] Press coverage of OpenAI investor communications (February 2026): infrastructure target reduced from ~$1.4T to ~$600B through 2030; ~$280B projected 2030 revenue from $13.1B in 2025; ~$8B 2025 cash burn; ~$100B round at ~$830-852B valuation.

[12] Nvidia quarterly results, FY2025-FY2027: datacenter revenue $115.2B (FY2025), $193.7B (FY2026), $75.2B (Q1 FY2027, total revenue $81.6B, +85% y/y).

[13] Hyperscaler Q4 2025 and Q1 2026 earnings disclosures: combined 2026 capex guidance ~$725B; Microsoft $190B (including ~$25B attributed to memory and component inflation); Google Cloud +63% y/y; Oracle remaining performance obligations ~half a trillion dollars.

[14] TSMC Q1 2026 results and investor commentary: HPC at 61% of revenue; 66.2% gross margin; 2026 capex $52-56B.

[15] Financial Times and CreditSights compilations of hyperscaler capex, AI shares (~75% for 2026), capital intensity (45-57%; Oracle 86%), and debt issuance ($108B in 2025; ~$1.5T projected); Morgan Stanley and Goldman Sachs capex projections as reported.

[16] TrendForce and trade-press reporting on TSMC CoWoS capacity (~35K wpm end-2024, ~75K end-2025, 125-130K target end-2026; 10-20% supply gap) and on memory: HBM share of DRAM wafer starts 8% (2024) to 23% (2026); ~3:1 HBM:DDR5 wafer intensity; Micron consumer-memory exit; DDR5 contract pricing +100%+.

[17] Econbrowser, "q-theory in a Time of AI" (December 2025), econbrowser.com - aggregate Tobin's q against the intellectual-property capital stock, Federal Reserve Z.1 data.

[18] A. Odlyzko, "Internet traffic growth: Sources and implications" (2003) and related papers on the doubling-every-100-days myth and telecom-bubble capital allocation.

[19] W. Nordhaus, "Schumpeterian Profits in the American Economy: Theory and Measurement," NBER Working Paper 10433 (2004).

[20] C. Perez, Technological Revolutions and Financial Capital (2002); W. Janeway, Doing Capitalism in the Innovation Economy (2012).

[21] The Economist, "The great telecoms crash" and "Too many debts; too few calls" (July 2002) - sector debt ~$1 trillion, up to half unrecoverable; M. Powell, testimony before the Senate Commerce Committee (July 30, 2002) - ~$2 trillion of market value and ~$1 trillion of debt lost, ~500,000 jobs (the $2T figure is Powell's, not The Economist's); contemporaneous reporting on WorldCom, Global Crossing, Level 3, and European 3G auctions.

[22] SEC v. Lucent Technologies Inc., Litigation Release 18715 (May 2004); contemporaneous reporting on Lucent vendor financing, the Winstar commitment and litigation, and the fiscal-2000 restatement.

[23] Nortel Networks bankruptcy proceedings (January 2009) and the Rockstar consortium patent auction ($4.5B, 2011), including Google's opening bid of $3.14159B; court records on estate professional fees and pension outcomes.

[24] CNBC reporting on the Situational Awareness unwind (D. Faber and others, July 30-31, 2026): forced sale of the public book to Citadel; margin calls from Bank of America, Goldman Sachs, and JPMorgan; peak ~$45B and residual ~$10B; retained Anthropic stake not for sale per spokesman; the Wall Street Journal's identification of Citadel; the Financial Times' report of the July 24 investor letter (439% net H1); Bloomberg on reopened capital raising. cnbc.com/2026/07/30 and /2026/07/31.

[25] SpotGamma, "Anatomy of a Margin Call: How Situational Awareness LP Unwound a $20 Billion AI Book in One Trade" (July 31, 2026) - unwind mechanics, disclosed positions, leverage, and the July 29 bottoms in core longs. spotgamma.com.

[26] Microsoft FQ4 2026 results and coverage (July 29-30, 2026): CNBC results report ($3.2B gain on Anthropic investment; Azure FY26 revenue above $100B, +41%; 30M+ paid Copilot seats; shares -19% YTD into the print; first voluntary retirement program); Bloomberg (shares +16%, fastest cloud growth in four years); Fortune ($480B single-session market-value gain); Investing.com (Azure +43% vs ~40% consensus; FQ4 capex $41B, ~two-thirds short-lived assets; guidance); CNBC/S3 Partners (July 28: 92M shares short, largest short interest since May 2015, no covering into earnings).

[27] Software-rotation coverage: Stocktwits (August 3, 2026) - July returns of Workday +31%, Adobe +22.1%, Intuit +21.6%, Autodesk +20.4%, Salesforce +17.5%, Microsoft +24.6%, on rotation out of chip stocks; 24/7 Wall St. (July 27, 2026) - YTD drawdowns (Salesforce -33%, Adobe -31%) and the "SaaSpocalypse" framing; CNBC (May 29, 2026) on the earlier software rebound from April lows.

[28] F. P. Stansberry / Porter & Co. (August 2026), essay on the Situational Awareness unwind and the toll-booth thesis, circulated on X and via members.porterandcompanyresearch.com - source, as relayed, of the per-name July long/short returns; hyperscaler capex-to-operating-cash-flow ratios; Moody's 2026-27 capex and debt projections; Bain and Sequoia (Cahn) revenue-gap estimates; the depreciation-extension inventory and Amazon's reversal; Oracle FY2026 and CoreWeave financing terms; the Meta Hyperion SPV terms; OpenAI audited 2025 results; the 1865-77 railroad statistics; and the express-company and Pullman financials. A marketing document for the author's newsletter; aggregate claims verified against [24]-[27] and historical claims against [29]-[31] where possible; the depreciation inventory, Moody's and Bain figures, and mileage/ton-mile series have since been re-anchored to primaries [33]-[36].

[29] US railroad history: R. Fogel, Railroads and American Economic Growth (1964); A. Fishlow (1965); Donaldson & Hornbeck, NBER WP 19213; Pereira, W&M economics WP 153 (antebellum returns); Depression of 1882-85 literature, including the estimate of railroads at ~15% of 1880s US capital formation; Investment Research Partners (August 2025) - datacenter capex ~1.2% of US GDP (2025) vs telecom ~1.0% (2000) vs railroads ~6.0% (1880s); Fabricated Knowledge, "Lessons from History: The Great Railroad Buildout" (December 2025) - capitalization $4.6B (1876) to $10.6B (1890), ~40% "water"; standard receivership accounts for 1873-77 and 1893-94.

[30] Railway Mania: G. Campbell, "Deriving the Railway Mania" (2013) and VoxEU/CEPR column (prices +106% 1843-45; dividends 4.4% to 7% to 2.4%; shares not obviously mispriced at peak); Campbell & Turner (2012; 2015); A. Odlyzko, "The collapse of the Railway Mania" and related; Bruner et al., Journal of Applied Corporate Finance (2025); Business History (2022-24) on managerial failure and the incumbents' expansion.

[31] Express companies and Pullman: standard histories (A. Harlow, Old Waybills; company records); Adams Express reorganization as a closed-end fund (1929; today Adams Diversified Equity Fund); Parcel Post Act (1913); wartime consolidation into American Railway Express (1918); Pullman antitrust consent decree and divestiture (1944-47). Specific dividend and capital figures as relayed by [28]; see Data Provenance.

[32] Credit Mobilier: standard accounts of the Union Pacific construction financing, the congressional share distribution, and the 1872-73 investigation and censures.

[33] Useful-life change-in-estimate disclosures, verified verbatim (August 2026): Microsoft 10-K FY2023 (+$3.7B operating income / +$3.0B net); Alphabet 10-K FY2023 (+$3.9B depreciation reduction / +$3.0B net); Amazon 10-K FY2024 (+$3.2B D&A / +$2.5B net) and 10-K FY2025 with Q1-2025 10-Q (the 6-to-5-year reversal: +$1.4B D&A / -$1.0B net in 2025, primarily AWS; separate from ~$920M Q4-2024 accelerated depreciation on early retirements); Meta 10-K FY2024 and FY2025 (+$2.92B depreciation / +$2.59B net); Oracle 10-K FY2025 (+$733M opex / +$573M net; an earlier FY2023 change added $434M). All at SEC EDGAR.

[34] Railroad primaries: Historical Statistics of the United States, Colonial Times to 1970 (Bicentennial Ed.), Series Q 321 (miles of road operated: 1865 = 35,085; 1875 = 74,096; first differences 2,665 in 1878 and 11,569 in 1882) and Q 329 (miles built: 7,439 in 1872) - note the construction figures mix the two adjacent series; both are Poor's-derived. NBER Macrohistory A0303BUSA259NNBR (revenue per freight ton-mile, all railroads, 1882-1911: 1900 = 0.729 cents) and A02F2AUSA374NNBR (miles built); A. Fishlow, "Productivity and Technological Change in the Railroad Sector, 1840-1910" (NBER, 1966), Table 1 (1870 = 2.18 cents, an estimate by his own caveat). Figures nominal; the general price level also fell 1870-1900.

[35] Moody's Ratings, hyperscaler capex forecast update (May 2026): $785B in 2026, approaching $1T in 2027, for the six largest US hyperscalers (Microsoft, Amazon, Meta, Alphabet, Oracle, CoreWeave); an $85B markup from the March 2026 forecast. As covered by Data Center Dynamics and CNBC (July 24, 2026). AWS capex is Moody's estimate. Distinct from Moody's December 2025 global datacenter outlook (>$3T over five years).

[36] Bain & Company, 6th annual Global Technology Report, press release (September 23, 2025): roughly $2 trillion in annual revenue needed by 2030 to fund AI compute demand; "even with AI-related savings, the world is still $800 billion short." bain.com.

[37] Inference margins and revenue mix (August 2026 verification pass): DeepSeek, "Overview of DeepSeek-V3/R1 Inference System" (open-infra-index, February 2025) - $87,072 daily GPU cost at an assumed $2/hr H800 against $562,027 theoretical daily revenue, self-labeled theoretical, with free web/app tiers and off-peak discounts excluded. OpenAI adjusted gross margin ~33% (2025) vs 40% (2024) and a 46% plan, and the ~70% figure as a compute margin rather than gross margin: The Information, as relayed in secondary coverage - treat as reported-not-disclosed. Anthropic gross-margin guidance cut ~10 points to ~40% on third-party inference overrun (~23% over plan): The Information, January 2026. Revenue mix: OpenAI API ~25% of a $12B run rate (Financial Times, August 2025), the majority being subscriptions; Anthropic ~75-85% API and enterprise (estimated, no segment disclosure - the S-1 is confidential). Application-layer gross margins 33% (2024) / 38% (2025) / 45% (2026 projected): ICONIQ survey of ~300 software executives, published January 2026; negative gross margins at AI-native coding firms and the recovery via backward integration, reported via investor sourcing. S. Altman, public statement (January 2025), on losing money at $200 per month.

Second edition, compiled August 4, 2026, from a working dialogue; first edition July 27, 2026. Estimates are the authors' and carry the uncertainties stated in the Data Provenance note; reported figures are as of their cited dates and may since have been revised.